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Solution to the equations of parallel transport in Kerr geometry; tidal tensor

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It is shown how the special separability properties of the Kerr solution can be used to obtain an explicit analytic solution to the problem of constructing an orthonormal tetrad that is parallel-propagated along an arbitrary time-like geodesic. The components of the tidal tensor are calculated explicitly in terms of this tetrad. The special case of geodesics lying in the equatorial plane is examined.

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1. Introduction

It is well known that the geodesic equation in the Kerr space–time can be solved analytically (Carter 1968*a*). The purpose of this paper is to show that it is also possible to solve analytically the equations of parallel transport applied to an orthonormal tetrad along an arbitrary time-like geodesic (Marck 1982). This work is presented in two steps in §3. Given any time-like geodesic, we first construct a standard locally defined orthonormal rest-frame tetrad along this geodesic with the property that two of its four vectors are already parallel-transported. We then obtain the desired entirely parallel-transported tetrad by means of a space rotation whose angle is shown to be the sum of two functions, one of coordinate r only, the other of θ only, given in terms of elliptic integrals.

This result will be useful for many problems concerning the behaviour of a body moving in the Kerr space–time. In particular it can facilitate the study of the tidal effect occurring in the neighbourhood of a Kerr black-hole (Mashhoon 1975; Stewart & Walker 1973). To facilitate a direct application to such problems, we compute explicitly in §4 the components of the tidal tensor by means of the tetrad constructed in §3.

The equatorial plane of the Kerr solution has been widely studied because of the simplifications that occur in it and because it is of great astrophysical interest owing to its symmetries. It has therefore been considered useful to present the results of §§3 and 4 in the special case of time-like geodesics lying in the equatorial plane, which is done in §5.

Although the results presented here are comparatively simple, they were obtained only after lengthy calculations, most of which were executed on computers with algebraic programs. Among these many were computed at the Centre Inter Regional de Calcul Electronique with the language A.M.P. (Drouffe 1975).

2. The Kerr metric and the symmetric tetrad

Throughout this paper all indices run from 0 to 3 except i and j, which run from 1 to 3. The indices in round brackets are tetrad indices and are used to define the components of any geometrical object by means of Carter's symmetric tetrad, which is introduced below. We shall use the Boyer-Lindquist (1967) coordinate system. The Kerr metric, with respect to this, takes the form

$$\mathrm{d}s^2 = -(1-2Mr/\Sigma)\,\mathrm{d}t^2 - 2(2Mr/\Sigma)a\sin^2\theta\,\mathrm{d}t\,\mathrm{d}\phi + (\Sigma/\Delta)\,\mathrm{d}r^2 + \Sigma\,\mathrm{d}\theta^2 + (\mathscr{A}/\Sigma)\sin^2\theta\,\mathrm{d}\phi^2 \quad (1)$$

where

$$\Sigma = r^2 + a^2\cos^2\theta, \quad (2)$$

$$\Delta = r^2 + a^2 - 2Mr, \quad (3)$$

$$\mathscr{A} = (r^2+a^2)^2 - \Delta\, a^2\sin^2\theta \quad (4)$$

in units such that $c = G = 1$.

The first integrals of the equations of motion (Carter 1968*a*) are

$$\dot{t} = (\mathscr{A}E - 2Mra\Phi)/\Delta\Sigma, \quad (5)$$

$$\Sigma^2\dot{r}^2 = \{E(r^2+a^2) - a\Phi\}^2 - \Delta(\mu^2 r^2 + K), \quad (6)$$

$$\Sigma^2\dot{\theta}^2 = K - \mu^2 a^2\cos^2\theta - (aE\sin\theta - \Phi/\sin\theta)^2, \quad (7)$$

$$\dot{\phi} = \frac{1}{\Delta}\left\{\frac{2Mr}{\Sigma}aE + \left(1-\frac{2Mr}{\Sigma}\right)\frac{\Phi}{\sin^2\theta}\right\} \quad (8)$$

where a dot denotes differentiation with respect to the proper time, τ say, and the constants of motion μ^2, E and Φ are respectively the rest mass, the energy and the angular momentum about the axis of symmetry of the orbiting test-particle, K being Carter's fourth constant.

In what follows, we shall consider only time-like geodesics, so we can normalize the proper time such that $\mu^2 = 1$.

It has been found convenient in carrying out calculations in the Kerr space–time (especially the calculations that are of interest here) to use the canonical symmetric orthonormal tetrad, ω say, introduced first by Carter (1968*b*), which has been shown to be the simplest for treating separability properties (Carter & McLenaghan 1982). It is given by

$$\omega^{(0)} = (\Delta/\Sigma)^{\frac{1}{2}}(\mathrm{d}t - a\sin^2\theta\,\mathrm{d}\phi), \quad (9)$$

$$\omega^{(1)} = (\Sigma/\Delta)^{\frac{1}{2}}\mathrm{d}r, \quad (10)$$

$$\omega^{(2)} = \Sigma^{\frac{1}{2}}\mathrm{d}\theta, \quad (11)$$

$$\omega^{(3)} = (\sin\theta/\Sigma^{\frac{1}{2}})\{a\,\mathrm{d}t - (r^2+a^2)\,\mathrm{d}\phi\}. \quad (12)$$

The metric tensor then takes the form

$$\mathrm{d}s^2 = \eta_{(a)(b)}\,\omega^{(a)}\omega^{(b)} \quad (13)$$

where $\eta = \mathrm{diag}(-1, +1, +1, +1)$ is the metric tensor of the Minkowski flat space–time.

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The Weyl tensor (in this case identical to the Riemann tensor) may be presented† in the form

$$\Omega^{(1)}{}_{(2)} = -I_1\omega^{(1)}\wedge\omega^{(2)}+I_2\omega^{(0)}\wedge\omega^{(3)}, \tag{14}$$

$$\Omega^{(0)}{}_{(3)} = -I_1\omega^{(0)}\wedge\omega^{(3)}-I_2\omega^{(1)}\wedge\omega^{(2)}, \tag{15}$$

$$\Omega^{(0)}{}_{(1)} = -2I_1\omega^{(1)}\wedge\omega^{(0)}+2I_2\omega^{(2)}\wedge\omega^{(3)}, \tag{16}$$

$$\Omega^{(3)}{}_{(2)} = -2I_1\omega^{(2)}\wedge\omega^{(3)}-2I_2\omega^{(1)}\wedge\omega^{(0)}, \tag{17}$$

$$\Omega^{(0)}{}_{(2)} = I_1\omega^{(2)}\wedge\omega^{(0)}+I_2\omega^{(1)}\wedge\omega^{(3)}, \tag{18}$$

$$\Omega^{(3)}{}_{(1)} = I_1\omega^{(1)}\wedge\omega^{(3)}-I_2\omega^{(2)}\wedge\omega^{(0)} \tag{19}$$

where $\Omega^{(a)}{}_{(b)} = \frac{1}{2}\mathbb{C}^{(a)}{}_{(b)(c)(d)}\omega^{(c)}\wedge\omega^{(d)}$ is the curvature 2-form, I_1 and I_2 being defined by

$$I_1 = (Mr/\Sigma^3)(r^2-3a^2\cos^2\theta), \tag{20}$$

$$I_2 = (Ma\cos\theta/\Sigma^3)(3r^2-a^2\cos^2\theta). \tag{21}$$

3. Standard and parallel-propagated tetrads

In this section, we shall construct an orthonormal tetrad that is parallel-transported along an arbitrary time-like geodesic, $\mathscr{C}$ say. Let $\boldsymbol{\lambda} = (\boldsymbol{\lambda}_0, \boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \boldsymbol{\lambda}_3)$ be the desired tetrad, where $\boldsymbol{\lambda}_0$ is a time-like vector and $\boldsymbol{\lambda}_i$, $i = 1, 2, 3$, are three space-like vectors. We shall of course choose $\boldsymbol{\lambda}_0$ to be the unit vector tangent to the geodesic, which is obviously parallel-transported, and so its components are

$$\lambda_0{}^{(0)} = (1/\Delta\Sigma)^{\frac{1}{2}}\{E(r^2+a^2)-a\Phi\}, \tag{22}$$

$$\lambda_0{}^{(1)} = (\Sigma/\Delta)^{\frac{1}{2}}\dot{r}, \tag{23}$$

$$\lambda_0{}^{(2)} = \Sigma^{\frac{1}{2}}\dot{\theta}, \tag{24}$$

$$\lambda_0{}^{(3)} = (1/\Sigma)^{\frac{1}{2}}(aE\sin\theta-\Phi/\sin\theta). \tag{25}$$

On the other hand there exists in the Kerr space–time (and more generally in all empty space–times of Petrov type D, except type IIIB (Collinson 1974)) a Killing–Yano tensor of order two, namely a skew symmetric tensor $f_{\mu\nu}$ satisfying the equation

$$\nabla_\rho f_{\mu\nu}+\nabla_\nu f_{\mu\rho} = 0. \tag{26}$$

This implies, as pointed out by Penrose (1973), that the vector $\boldsymbol{L}$ defined by

$$L^\mu = f^\mu{}_\nu\lambda_0{}^\nu \tag{27}$$

is parallel-propagated along the geodesic $\mathscr{C}$ and orthogonal to $\boldsymbol{\lambda}_0$.

For the Kerr solution, the tetrad components of the Killing–Yano tensor (up to a constant multiplicative factor) can be expressed as

$$f_{(a)(b)}\omega^{(a)}\wedge\omega^{(b)} = -r\omega^{(2)}\wedge\omega^{(3)}+a\cos\theta\omega^{(1)}\wedge\omega^{(0)}. \tag{28}$$

† This version rectifies several sign errors in the expression originally given by Carter (1973).

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Using these results and the fact that $L^{(a)}L_{(a)} = K$, we can choose $\boldsymbol{\lambda}_2$ to be the parallel-propagated unit vector $(1/K)^{\frac{1}{2}}\boldsymbol{L}$, which is orthogonal to $\boldsymbol{\lambda}_0$ and whose components are

$$\lambda_2{}^{(0)} = (\Sigma/K\Delta)^{\frac{1}{2}}\, a\cos\theta\dot{r}, \tag{29}$$

$$\lambda_2{}^{(1)} = (1/K\Sigma\Delta)^{\frac{1}{2}}\, a\cos\theta\{E(r^2+a^2) - a\Phi\}, \tag{30}$$

$$\lambda_2{}^{(2)} = -(1/K\Sigma)^{\frac{1}{2}}\, r(aE\sin\theta - \Phi/\sin\theta), \tag{31}$$

$$\lambda_2{}^{(3)} = (\Sigma/K)^{\frac{1}{2}}\, r\dot{\theta}. \tag{32}$$

At this point it can be seen that there is a particularly natural standard choice of two more vectors $\tilde{\lambda}_1$ and $\tilde{\lambda}_3$ that form a complete orthonormal basis when taken in conjunction with $\boldsymbol{\lambda}_0$ and $\boldsymbol{\lambda}_2$ as defined above:

$$\tilde{\lambda}_1{}^{(0)} = \alpha(\Sigma/K\Delta)^{\frac{1}{2}}\, r\dot{r}, \tag{33}$$

$$\tilde{\lambda}_1{}^{(1)} = \alpha(1/K\Sigma\Delta)^{\frac{1}{2}}\, r\{E(r^2+a^2) - a\Phi\}, \tag{34}$$

$$\tilde{\lambda}_1{}^{(2)} = \beta(1/K\Sigma)^{\frac{1}{2}}\, a\cos\theta(aE\sin\theta - \Phi/\sin\theta), \tag{35}$$

$$\tilde{\lambda}_1{}^{(3)} = -\beta(\Sigma/K)^{\frac{1}{2}}\, a\cos\theta\dot{\theta} \tag{36}$$

and

$$\tilde{\lambda}_3{}^{(0)} = \alpha(1/\Sigma\Delta)^{\frac{1}{2}}\{E(r^2+a^2) - a\Phi\}, \tag{37}$$

$$\tilde{\lambda}_3{}^{(1)} = \alpha(\Sigma/\Delta)^{\frac{1}{2}}\dot{r}, \tag{38}$$

$$\tilde{\lambda}_3{}^{(2)} = \beta\Sigma^{\frac{1}{2}}\dot{\theta}, \tag{39}$$

$$\tilde{\lambda}_3{}^{(3)} = \beta(1/\Sigma)^{\frac{1}{2}}(aE\sin\theta - \Phi/\sin\theta) \tag{40}$$

where

$$\alpha = (K - a^2\cos^2\theta)^{\frac{1}{2}}/(r^2+K)^{\frac{1}{2}}, \tag{41}$$

$$\beta = (r^2+K)^{\frac{1}{2}}/(K - a^2\cos^2\theta)^{\frac{1}{2}}. \tag{42}$$

The standard locally defined tetrad set up in this way is not quite the desired result since unlike $\boldsymbol{\lambda}_0$ and $\boldsymbol{\lambda}_2$ the canonically chosen vectors $\tilde{\boldsymbol{\lambda}}_1$ and $\tilde{\boldsymbol{\lambda}}_3$ are not parallel-propagated. However, they can be related by a single time-dependent rotation angle Ψ to two new orthonormal unit vectors $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_3$ chosen in the same plane in such a way that the required property

$$\lambda_0{}^{\mu}\nabla_{\mu}\lambda_k{}^{\nu} = 0 \tag{43}$$

is satisfied not only for $k = 0$ and 2 but also for $k = 1$ and 3.

Introducing the angle Ψ explicitly by

$$\boldsymbol{\lambda}_1 = \tilde{\boldsymbol{\lambda}}_1\cos\Psi - \tilde{\boldsymbol{\lambda}}_3\sin\Psi \tag{44}$$

and

$$\boldsymbol{\lambda}_3 = \tilde{\boldsymbol{\lambda}}_1\sin\Psi + \tilde{\boldsymbol{\lambda}}_3\cos\Psi \tag{45}$$

one finds after a lengthy calculation that the equation governing the variations of Ψ can be expressed in the following form:

$$\dot{\Psi} = \frac{K^{\frac{1}{2}}}{\Sigma}\left\{\frac{E(r^2+a^2) - a\Phi}{r^2+K} + a\,\frac{(\Phi - aE\sin^2\theta)}{K - a^2\cos^2\theta}\right\}. \tag{46}$$

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It is easy to show by making use of equations (6) and (7) that Ψ can be written as

$$\Psi = F(r) + G(\theta) \tag{47}$$

where F and G are respectively functions of r and θ only defined by

$$F(r) = \epsilon_1 K^{\frac{1}{2}} \int^r \frac{E(r^2+a^2) - a\Phi}{r^2+K} \frac{\mathrm{d}r}{[\{E(r^2+a^2) - a\Phi\}^2 - \Delta)r^2 + K)]^{\frac{1}{2}}}, \tag{48}$$

$$G(\theta) = \epsilon_2 a K^{\frac{1}{2}} \int^\theta \frac{\Phi - aE\sin^2\theta}{(K - a^2\cos^2\theta)} \frac{\mathrm{d}\theta}{\{K - a^2\cos^2\theta - (aE\sin\theta - \Phi/\sin\theta)^2\}^{\frac{1}{2}}}, \tag{49}$$

ϵ_1 being the sign of $\dot{r}$ and ϵ_2 the sign of $\dot{\theta}$. It is to be noticed that in the limit $M \to 0$ these formulae are related by the algebraic symmetry modulo analytic continuation between r and $\cos\theta$ in that one has $G(\theta) = \pm F(ia\cos\theta)$.

4. Tidal tensor

The most convenient way to calculate the relative acceleration (tidal effect) of test-particles moving in the neighbourhood of a time-like geodesic is by evaluating the Weyl tensor in some appropriate parallel-propagated orthonormal tetrad $(\boldsymbol{\lambda}_0, \boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \boldsymbol{\lambda}_3)$ defined along this geodesic, $\boldsymbol{\lambda}_0$ being the four-velocity. As shown by Pirani (1956) we need only the tidal tensor components

$$C_{ij} = \mathbb{C}_{(a)(b)(c)(d)} \lambda_0^{(a)} \lambda_i^{(b)} \lambda_0^{(c)} \lambda_j^{(d)}. \tag{50}$$

From the well known properties of the Weyl tensor we can see that the tidal tensor takes the form of a symmetric, traceless, 3×3 matrix.

Using the expression for the Weyl tensor given by equations (14)–(19) we can compute the components of the tidal tensor in the Kerr geometry. For any orthonormal tetrad $\boldsymbol{\lambda}$ we get for $i \neq j$

$$\begin{aligned} C_{ij} = {} & 3\{(\lambda_0^{(0)}\lambda_i^{(1)} - \lambda_0^{(1)}\lambda_i^{(0)})(\lambda_0^{(1)}\lambda_j^{(0)} - \lambda_0^{(0)}\lambda_i^{(1)}) \\ & + (\lambda_0^{(2)}\lambda_i^{(3)} - \lambda_0^{(3)}\lambda_i^{(2)})(\lambda_0^{(2)}\lambda_j^{(3)} - \lambda_0^{(3)}\lambda_j^{(2)})\} I_1 \\ & - 3\{(\lambda_0^{(0)}\lambda_j^{(1)} - \lambda_0^{(1)}\lambda_j^{(0)})(\lambda_0^{(2)}\lambda_i^{(3)} - \lambda_0^{(3)}\lambda_i^{(2)}) \\ & + (\lambda_0^{(0)}\lambda_i^{(1)} - \lambda_0^{(1)}\lambda_i^{(0)})(\lambda_0^{(2)}\lambda_j^{(3)} - \lambda_0^{(3)}\lambda_j^{(2)})\} I_2, \end{aligned} \tag{51}$$

and for $i = j$

$$\begin{aligned} C_{ii} = {} & [1 - 3\{(\lambda_0^{(0)}\lambda_i^{(1)} - \lambda_0^{(1)}\lambda_i^{(0)})^2 - (\lambda_0^{(2)}\lambda_i^{(3)} - \lambda_0^{(3)}\lambda_i^{(2)})^2\}] I_1 \\ & - 6(\lambda_0^{(0)}\lambda_i^{(1)} - \lambda_0^{(1)}\lambda_i^{(0)})(\lambda_0^{(2)}\lambda_i^{(3)} - \lambda_0^{(3)}\lambda_i^{(2)}) I_2 \end{aligned} \tag{52}$$

where the quantities I_1 and I_2 are defined by equations (20) and (21).

In particular, if $\boldsymbol{\lambda}$ is now defined as in § 3, we can explicitly evaluate the components of the tidal tensor. We obtain the following result:

$$C_{11} = \left\{1 - 3\frac{ST(r^2 - a^2\cos^2\theta)}{K\Sigma^2}\cos^2\Psi\right\} I_1 + 6ar\cos\theta\frac{ST}{K\Sigma^2}\cos^2\Psi I_2, \tag{53}$$

$$C_{12} = \{-ar\cos\theta(S+T) I_1 + (a^2\cos^2\theta S - r^2 T) I_2\} 3\frac{(ST)^{\frac{1}{2}}}{K\Sigma^2}\cos\Psi, \tag{54}$$

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$$C_{13} = \{(a^2\cos^2\theta - r^2)I_1 + 2ar\cos\theta I_2\}\,3\frac{ST}{K\Sigma^2}\cos\Psi\sin\Psi, \tag{55}$$

$$C_{22} = \left(1 + 3\frac{r^2T^2 - a^2\cos^2\theta S^2}{K\Sigma^2}\right)I_1 - 6ar\cos\theta\frac{ST}{K\Sigma^2}I_2, \tag{56}$$

$$C_{23} = \{-ar\cos\theta(S+T)I_1 + (a^2\cos^2\theta S - r^2T)I_2\}\,3\frac{(ST)^{\frac{1}{2}}}{K\Sigma^2}\sin\Psi, \tag{57}$$

$$C_{33} = \left\{1 - 3\frac{ST(r^2 - a^2\cos^2\theta)}{K\Sigma^2}\sin^2\Psi\right\}I_1 + 6ar\cos\theta\frac{ST}{K\Sigma^2}\sin^2\Psi I_2 \tag{58}$$

where

$$S = r^2 + K \tag{59}$$

and

$$T = K - a^2\cos^2\theta. \tag{60}$$

5. Equatorial plane

In this section we shall present the results obtained from §§3 and 4 in the particular case of geodesic motion in the equatorial plane, i.e. when the coordinate $\theta = \frac{1}{2}\pi$ and $\dot{\theta} = 0$ and the parameters a, E, Φ and K satisfy the relation

$$K = (aE - \Phi)^2. \tag{61}$$

With use of the Kronecker symbol $\delta_\nu{}^\mu$, the parallel-propagated tetrad $\boldsymbol{\lambda}$ takes the form

$$\lambda_0{}^{(\mu)} = (1/r\Delta^{\frac{1}{2}})\{E(r^2+a^2) - a\Phi\}\,\delta_0{}^\mu + (r\dot{r}/\Delta^{\frac{1}{2}})\,\delta_1{}^\mu + (1/r)\,(aE - \Phi)\,\delta_3{}^\mu, \tag{62}$$

$$\lambda_1{}^{(\mu)} = \tilde{\lambda}_1{}^{(\mu)}\cos\Psi - \tilde{\lambda}_3{}^{(\mu)}\sin\Psi, \tag{63}$$

$$\lambda_2{}^{(\mu)} = \{(\Phi - aE)/K^{\frac{1}{2}}\}\,\delta_2{}^\mu, \tag{64}$$

$$\lambda_3{}^{(\mu)} = \tilde{\lambda}_1{}^{(\mu)}\sin\Psi + \tilde{\lambda}_1{}^{(\mu)}\cos\Psi \tag{65}$$

where the standard tetrad vectors $\tilde{\boldsymbol{\lambda}}_1$, $\tilde{\boldsymbol{\lambda}}_2$ are given by

$$\tilde{\lambda}_1{}^{(\mu)} = \frac{r^2\dot{r}}{\{\Delta(r^2+K)\}^{\frac{1}{2}}}\,\delta_0{}^\mu + \frac{E(r^2+a^2) - a\Phi}{\{\Delta(r^2+K)\}^{\frac{1}{2}}}\,\delta_1{}^\mu \tag{66}$$

and

$$\tilde{\lambda}_3{}^{(\mu)} = \frac{K^{\frac{1}{2}}\{E(r^2+a^2) - a\Phi\}}{r\Delta^{\frac{1}{2}}(r^2+K)^{\frac{1}{2}}}\,\delta_0{}^\mu + \frac{K^{\frac{1}{2}}r\dot{r}}{\Delta^{\frac{1}{2}}(r^2+K)^{\frac{1}{2}}}\,\delta_1{}^\mu + \frac{(r^2+K)\,(aE - \Phi)}{rK^{\frac{1}{2}}}\,\delta_3{}^\mu. \tag{67}$$

Then Ψ (function only of r) is determined by

$$\dot{\Psi} = A/(r^2+K) \tag{68}$$

where A is a constant defined in terms of the parameters a, E, Φ and K as

$$A = EK^{\frac{1}{2}} - aK^{\frac{1}{2}}/(aE - \Phi). \tag{69}$$

We remark at this stage that the vector $\boldsymbol{\lambda}_2$, which is automatically parallel-transported, reduces in this case to a unit vector orthogonal to the equatorial plane.

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This is a generalization of the law of conservation of angular momentum in flat space–time.

The tidal tensor takes a very simple form in this case: the diagonal components are given explicitly by

$$C_{11} = \left\{1 - 3\frac{(r^2+K)}{r^2}\cos^2\Psi\right\}\frac{M}{r^3}, \tag{70}$$

$$C_{22} = \left(1 + 3\frac{K}{r^2}\right)\frac{M}{r^3}, \tag{71}$$

$$C_{33} = \left\{1 - 3\frac{(r^2+K)}{r^2}\sin^2\Psi\right\}\frac{M}{r^3} \tag{72}$$

and the only independent non-zero off-diagonal component is C_{13}, which can be written as

$$C_{13} = -3\frac{(r^2+K)}{r^5}M\cos\Psi\sin\Psi. \tag{73}$$

6. Conclusion

In a general space–time the equations of parallel transport may be very difficult to solve explicitly even along known geodesics. However, in the particular case of the Kerr solution we have shown that the special separability properties that permit the explicit integration of the geodesic equations can also be used to provide an explicit solution to the problem of constructing a parallel-transported orthonormal tetrad, $\boldsymbol{\lambda}$. This tetrad, which is given in terms of functions of the coordinates, allowed us to compute explicitly the tidal tensor.

Moreover, it is easy to show that this tetrad $\boldsymbol{\lambda}$ and the expressions (equations (53)–(58)) for the tidal tensor can be generalized to geodesic motion in the Kerr–Newman space–time (i.e. trajectories of test particles with zero electric charge). This is done by giving to the quantities Δ, I_1 and I_2 the following values:

$$\Delta = r^2 + a^2 - 2Mr + Q^2 + P^2, \tag{74}$$

$$I_1 = (1/\Sigma^3)\{Mr(r^2 - 3a^2\cos^2\theta) - (r^2 - a^2\cos^2\theta)(Q^2+P^2)\}, \tag{75}$$

$$I_2 = (1/\Sigma^3)\{Ma\cos\theta(3r^2 - a^2\cos^2\theta) - 2ar\cos\theta(Q^2+P^2)\} \tag{76}$$

where the parameters P and Q represent magnetic and electric monopole moments.

These results will be used in the study of the tidal effects for an affine star model (Carter & Luminet 1982) in the neighbourhood of a Kerr black hole.

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References

Boyer, R. H. & Lindquist, R. W. 1967 *J. math. Phys.* **8**, 265.
Carter, B. 1968*a* *Phys. Rev.* **174**, 1559.
Carter, B. 1968*b* *Communs math. Phys.* **10**, 280.
Carter, B. 1973 In *Black holes* (ed. C. de Witt & B. de Witt). New York: Gordon & Breach.
Carter, B. & Luminet, J.-P. 1982 *Nature, Lond.* **296**, 211.
Carter, B. & McLenaghan, R. G. 1982 In *Recent developments of general relativity* (2nd Marcel Grossman Meeting, Trieste 1979) (ed. R. Ruffini), p. 575. Amsterdam: North Holland.
Collinson, C. D. 1974 *Tensor* N.S. **28**, 173.
Drouffe, J.-M. 1975 AMP reference manual, CEN-Saclay.
Marck, J.-A. 1982 *C.r. hebd. Séanc. Acad. Sci., Paris* I **295**, 185.
Mashhoon, B. 1975 *Astrophys. J.* **197**, 705.
Penrose, R. 1973 *Ann. N.Y. Acad. Sci.* **224**, 125.
Pirani, F. A. E. 1956 *Acta phys. pol.* **15**, 389.
Stewart, J. & Walker, M. 1973 In *Springer tracts in modern physics* (ed. G. Höhler), p. 69. Berlin: Springer-Verlag.

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